

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [150, 103, 96, 113, 101, 108, 77, 116, 101, 96]

Tap amplitudes (pixels): [144.80331487918363, 128.37834708392222, 126.62543188475212, 110.01818031580054, 137.82960494755835, 134.23859355639868, 128.3627671873741, 124.77579893553076, 119.40686747419514, 127.45587471748802, 113.63538181394033]

GPT-5

MDS-UPDRS Item 3.4 Finger Tapping: Scoring Analysis

Given features

- Number of taps: 11 (10 intervals provided, implying 11 tap events were detected)
- Tap periods (frames): [150, 103, 96, 113, 101, 108, 77, 116, 101, 96]
- Tap amplitudes (pixels): [144.80, 128.38, 126.63, 110.02, 137.83, 134.24, 128.36, 124.78, 119.41, 127.46, 113.64]

Note: The task targets 10 taps. The data includes 11 amplitude samples (likely including initial touch), and 10 periods (consistent with 10 inter-tap intervals). I analyze taps 1–10 primarily, with tap 11 as the final outcome point for amplitude trend.

Speed (periods)

- Periods mostly between 96–116 frames, with one longer initial (150) and one faster (77).
- After the first interval, cadence is fairly consistent with modest variability; there is no evidence of marked slowing or sustained bradykinesia.
- No clear “moderate slowing”; slight slowing relative to initial? Hard to assert without baseline, but overall speed appears near steady with natural variability.

Interruptions/hesitations (rhythm)

- Interruptions/hesitations would show as unusually long periods or arrests. The first interval (150) could reflect a start-up effect; otherwise, there are no multiple long pauses. The shortest (77) is faster, not a hesitation.
- Counting interruptions: I do not see 3–5 clear hesitations nor >5 interruptions, and no long arrest/freezing.

Amplitude trend

- Amplitudes: 144.8 (tap1) → 128.4 → 126.6 → 110.0 → 137.8 → 134.2 → 128.4 → 124.8 → 119.4 → 127.5 → 113.6 (tap11).
- There is a notable drop by tap 4, partial recovery (taps 5–6), then a gradual decline with small recovery at tap 10, and a lower value at tap 11.
- Pattern shows some decrement beginning early (by tap 3–4), but not a monotonic progressive decrement from the start; rather, it dips then partially recovers and fluctuates.
- According to criteria:
 - “Decrements near the end” → aligns with slight (1).
 - “Decrements midway” → would align with mild (2); here, a dip occurs by tap 3–4 (early/mid), but recovery suggests not a sustained mid-sequence decrement.
 - “Decrements starting after the 1st tap” (3) implies a consistent progressive reduction after tap 1, which is not strictly observed due to recovery at taps 5–6 and 10.

Overall interpretation

- Speed: near-stable; no moderate slowing.
- Rhythm: no evidence of 3–5 interruptions or a long arrest.
- Amplitude: some variability with early dip and partial recovery; clearer lower amplitudes toward the end relative to the start, but not a consistent early-on progressive decrement.

Given these, the findings best fit “Slight” (1), particularly due to:

- Slight rhythm irregularity without multiple interruptions.
- No clear moderate slowing.
- Amplitude decrement more apparent toward the end rather than sustained from midway onward.

Final score: 1
